

Tooling release checklist

The questions to answer before you cut steel. After release, a change is measured in weeks and tens of thousands of euros.

Before you release hard tooling

- ☐ **Thermal** has been simulated and then measured on real hardware under worst-case conditions.
- ☐ **Tolerances** in the drawings can be held by a real production process (checked with the mould maker).
- ☐ The **assembly sequence** was walked through with the factory, step by step, with no manual rescue steps.
- ☐ Every **component** in the design can be sourced at volume, not just for prototypes.
- ☐ **EMC** pre-scan has been done on hardware that matches the final layout.
- ☐ Fit and function are proven on **production-intent** parts, not hand-made ones.
- ☐ The design is **frozen**: changes after this point need written approval and a cost estimate.
- ☐ The **cost of a change** after release is written down and signed off by the person paying.
- ☐ A named person has signed: "I would stake my own budget on this release."

Examples from the book

What was skipped	What it cost
Thermal was not resolved before committing to hard tooling.	A €80,000 tooling modification nobody had planned for.
Tolerances were set too tight in CAD, which looked fine on screen.	Specialised tooling in production; about €40k of re-work found at DVT.
Assembly sequence was not reviewed with the factory.	Manual intervention on 40% of units, invisible on a bench, a crisis at 10,000 units per week.
Component chosen because it was available for prototypes.	Single-source risk, locked in before anyone noticed.

Decision

Released?	Date	Approved by	Open issues accepted (with cost of leaving them)

Why: steel can be cut away but never put back. Releasing tooling because the schedule says so, instead of because the evidence says so, is the most expensive habit in hardware.